

James Kim

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EDUCATION

University of Southern California (USC)

Aug 2025 - Aug 2026

Masters in Aerospace Engineering

GPA 3.5 / 4.0

Relevant Coursework: Linear and Nonlinear Controls, Fundamentals in Combustion, Flight Mechanics, and Spacecraft attitude.

University of California, Los Angeles (UCLA)

June 2020 - June 2024

Bachelors in Biochemistry, Microbiology, Immunology, and Molecular Genetics (MIMG), and Neuroscience

GPA 3.67 / 4.0

PROFESSIONAL EXPERIENCE

GNC Simulations team member | USC Rocket Propulsion Laboratory

Aug 2025 - Aug 2026

- Analyzed flight recovery data from launch day using MATLAB to evaluate trajectory behavior, recovery performance, and flight system outputs. Performed further simulations to process previous troubleshooting errors and made corrections.
- Optimized the code for AimOn, a personalized software tool used for trajectory and flight performance analysis.
- Verified simulation outputs through Flighton, a personalized software tool that measures recovered flight data and identifies discrepancies in flight performance predictions.
- Improved accuracy of simulating Hovmöller diagrams used to predict wind response prior to launch days.

Energy Storage for EAP | NASA L'SPACE Workforce Experience

Aug 2024 - Mar 2025 (Remote)

- Lead Engineer of the NASA Proposal Writing and Evaluation Experience (NPWEE) project focused on creating an energy storage for electrified aircraft propulsion systems (EAP).
- Collaborated with a variety of engineering students from different Universities in America and submitted a written proposal on a hybrid system supporting proprietary aircraft energy architecture.
- Used Siemens NX for CAD design to draw diagrams for the written proposal. Technical design details are omitted due to NDA restrictions.

RESEARCH EXPERIENCE

Graduate Research Assistant | Center of Advanced Manufacturing, USC

Aug 2025 - Aug 2026

- Developed and integrated a camera-based motion tracking system using Intel RealSense depth sensing and ROS to measure 3D displacement for a physics-informed digital twin.
- Created calibration and validation procedures to compare camera-based displacement measurements against controlled ground-truth references, improving reliability of the hardware / software data pipeline.
- Fabricated and printed liquid metal sensors using gallium and indium. The sensors were tested for connectivity via bluetooth software.
- Molded 3D shaped elastomers used to attach sensors to measure displacement and ground-truth values.

Research Assistant | Serotonin Lab, UCLA

June 2021 - Jan 2023

- Analyzed real time simultaneous electrochemical sensor signals of Dopamine and Serotonin quantitatively through MATLAB.
- Through personalized software, optimized further signal processing in different control environments and debugged MATLAB error discrepancies from one environment to another.

SKILLS

GNC & Controls: State Space Modeling, Controllability & Observability, State Estimation, Kalman Filtering, Pole Placement, LQR, Observer Based Control, Lyapunov Stability, Feedback Linearization, Backstepping, Sliding Mode Control, Control Allocation

Flight Dynamics & Vehicle Modeling: 6-DOF Simulation, Translational & Rotational Dynamics, Reference Frames & Coordinate Transformations, DCMs, Quaternions, Trim & Linearization, Static and Dynamic Stability, Modal Analysis

Programming & Simulation: MATLAB, Simulink, Python, C++, Control System Toolbox, GitHub, Linux

Systems Engineering & V&V: System Integration & Test, Verification & Validation (V&V), Sensor Calibration, Test Procedure Development, Performance Analysis, Technical Documentation

Combustion & Propulsion: Chemical Kinetics, Detonation Wave Analysis, Rankine-Hugoniot & Rayleigh-Line Analysis, Chapman-Jouguet Condition, ZND Detonation-Wave Model

Engineering Tools: CATIA, JIRA, Siemens NX